Hardy-Rogers (Ćirić) Condition

$a_i \in [0,1)$ for $i = 1, 2, 3, 4, 5$

$a_1 + a_2 + a_3 + a_4 + a_5 < 1$

Contraction Formula:

$$d(Tx, Ty) \leq a_1 d(x, Tx) + a_2 d(y, Ty)$$

$$+ a_3 d(x, Ty) + a_4 d(y, Tx) + a_5 d(x, y)$$

Example Parameter Values

Banach: $a_5 = 0.5$, others = 0

Kannan: $a_1 = a_2 = 0.5$, others = 0

Chatterjea: $a_3 = a_4 = 0.5$, others = 0

Zamf. Special: $a_1 = a_2 = a_3 = a_4 = 0.25$

General: All $a_i > 0$ with sum < 1